



MOTOROLA

MC12040

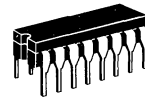
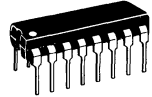
PHASE-FREQUENCY DETECTOR

The MC12040 is a phase-frequency detector intended for use in systems requiring zero phase and frequency difference at lock. In combination with a voltage controlled oscillator (such as the MC1648), it is useful in a broad range of phase-locked loop applications. Operation of this device is identical to that of Phase Detector #1 of the MC4044. A discussion of the theory of operation and applications information is given on the MC4344/4044 data sheet.

Operating Frequency = 80 MHz typical

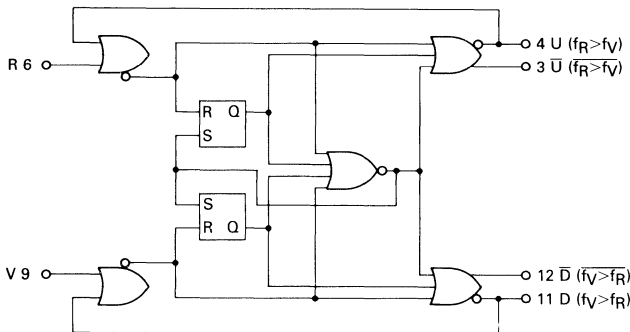
PHASE-FREQUENCY DETECTOR

P SUFFIX
PLASTIC PACKAGE
CASE 646



L SUFFIX
CERAMIC PACKAGE
CASE 632

LOGIC DIAGRAM



V_{CC1} = Pin 1
 V_{CC2} = Pin 14
 V_{EE} = Pin 7

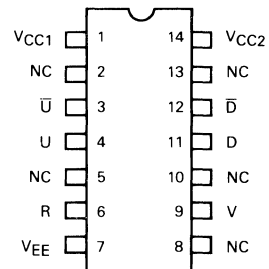
TRUTH TABLE

This is not strictly a functional truth table; i.e., it does not cover all possible modes of operation. However it gives a sufficient number of tests to ensure that the device will function properly in all modes of operation.

INPUT		OUTPUT				
R	V	U	D	U-bar	D-bar	
0	0	X	X	X	X	
0	1	X	X	X	X	
1	1	X	X	X	X	
0	1	X	X	X	X	
1	1	1	0	0	1	
0	1	1	0	0	1	
1	1	1	0	0	1	
1	0	1	0	0	1	
1	1	0	0	1	1	
1	0	0	0	1	1	
1	1	0	1	1	0	
1	0	0	1	1	0	
1	1	0	1	1	0	
0	1	0	1	1	0	
1	1	0	0	1	1	

X = Don't Care

PIN ASSIGNMENT

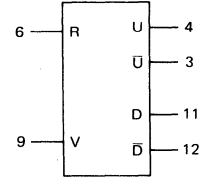


NC — No Connection

MC12040

ELECTRICAL CHARACTERISTICS

The MC12040 has been designed to meet the dc specifications shown in the test table after thermal equilibrium has been established. Outputs are terminated through a 50 ohm resistor to +3.0 V for +5.0 V tests and through a 50 ohm resistor to -2.0 V for -5.2 V tests.



Supply Voltage = -5.2V

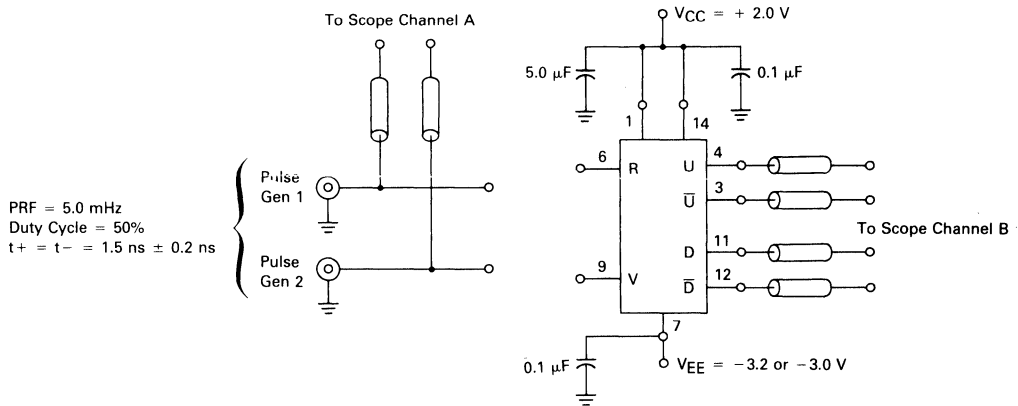
										TEST VOLTAGE VALUES (Volts)					(V _{CC}) Gnd
										V _{IH} max	V _{IL} min	V _{IHA} min	V _{ILA} max	V _{EE}	
										0°C	25°C	50°C	75°C	100°C	
										0°C	25°C	50°C	75°C	100°C	
										0°C	25°C	50°C	75°C	100°C	
Characteristic	Symbol	Pin Under Test	Min	Max	Min	Max	Min	Max	Unit	V _{IH} max	V _{IL} min	V _{IHA} min	V _{ILA} max	V _{EE}	
Power Supply Drain Current	I _E	7			-120	-60			mAdc	6	9			7	1,14
Input Current	I _{INH}	6 9				350 350			μAdc μAdc					7	1,14 1,14
Logic "1" Output Voltage	V _{OH} ①	3 4 11 12	-1.000	-0.840	-0.960	-0.810	-0.900	-0.720	Vdc					7	1,14
Logic "0" Output Voltage	V _{OL} ①	3 4 11 12	-1.870	-1.635	-1.850	-1.620	-1.830	-1.595	Vdc					7	1,14
Logic "1" Threshold Voltage	V _{OHA} ②	3 4 11 12	-1.020		-0.980		-0.920		Vdc			6,9		7	1,14
Logic "0" Threshold Voltage	V _{OLA} ②	3 4 11 12		-1.615		-1.600		-1.575	Vdc			9 6 9 6	6 9 6 9	7	1,14

Supply Voltage = +5.0V

										TEST VOLTAGE VALUES (Volts)					(V _{EE}) Gnd
										V _{IH} max	V _{IL} min	V _{IHA} min	V _{ILA} max	V _{CC}	
										0°C	25°C	50°C	75°C	100°C	
										0°C	25°C	50°C	75°C	100°C	
										0°C	25°C	50°C	75°C	100°C	
Characteristic	Symbol	Pin Under Test	Min	Max	Min	Max	Min	Max	Unit	V _{IH} max	V _{IL} min	V _{IHA} min	V _{ILA} max	V _{CC}	
Power Supply Drain Current	I _E	7			-115	-60			mAdc	6	9			1,14	7
Input Current	I _{INH}	6 9				350 350			μAdc μAdc					1,14 1,14	7
Logic "1" Output Voltage	V _{OH} ①	3 4 11 12	4.000	4.160	4.040	4.190	4.100	4.280	Vdc					1,14	7
Logic "0" Output Voltage	V _{OL} ①	3 4 11 12	3.190	3.430	3.210	3.440	3.230	3.470	Vdc					1,14	7
Logic "1" Threshold Voltage	V _{OHA} ②	3 4 11 12	3.980		4.020		4.080		Vdc			6,9		1,14	7
Logic "0" Threshold Voltage	V _{OLA} ②	3 4 11 12		3.450		3.460		3.490	Vdc			9 6 9 6	6 9 6 9	1,14	7

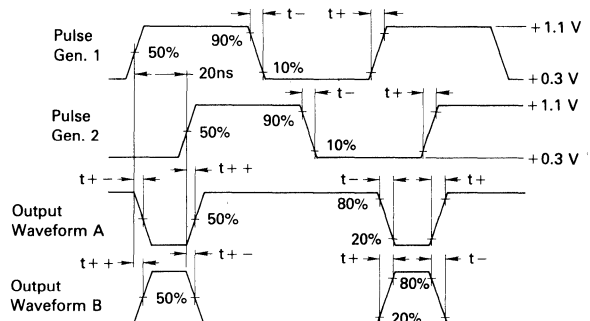
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AC TESTS



NOTES:

1. All input and output cables to the scope are equal lengths of 50 Ω coaxial cable.
2. Unused input and outputs are connected to a 50 Ω resistor to ground.
3. The device under test must be preconditioned before performing the ac tests. Preconditioning may be accomplished by applying pulse generator 1 for a minimum of two pulses prior to pulse generator 2. The device must be preconditioned again when inputs to pins 6 and 9 are interchanged. The same technique applies.



Characteristic	Symbol	Pin Under Test	Output Waveform	MC12040			Unit	TEST VOLTAGES/WAVEFORMS APPLIED TO PINS LISTED BELOW:			
				0°C	+25°C	+75°C		Pulse Gen. 1	Pulse Gen. 2	VEE -3.0 or -3.2 V	VCC +2.0 V
				Max	Max	Max					
Propagation Delay	t_{6+4+}	6,4	B	4.6	4.6	5.0	ns	6	9	7	1,14
	t_{6+12+}	6,12	A	6.0	6.0	6.6		9	6		
	t_{6+3-}	6,3	A	4.5	4.5	4.9		6	9		
	t_{6+11-}	6,11	B	6.4	6.4	7.0		9	6		
	t_{9+11+}	9,11	B	4.6	4.6	5.0		9	6		
	t_{9+3+}	9,3	A	6.0	6.0	6.6		6	9		
	t_{9+12-}	9,12	A	4.5	4.5	4.9		9	6		
	t_{9+4-}	9,4	B	6.4	6.4	7.0		6	9		
Output Rise Time	t_{3+}	3	A	3.4	3.4	3.8	ns	6	9	7	1,14
	t_{4+}	4	B					6	9		
	t_{11+}	11	B					9	6		
	t_{12+}	12	A					9	6		
Output Fall Time	t_{3-}	3	A	3.4	3.4	3.8	ns	6	9	7	1,14
	t_{4-}	4	B					6	9		
	t_{11-}	11	B					9	6		
	t_{12-}	12	A					9	6		

APPLICATIONS INFORMATION

The MC12040 is a logic network designed for use as a phase comparator for MECL-compatible input signals. It determines the "lead" or "lag" phase relationship and the time difference between the leading edges of the waveforms. Since these edges occur only once per cycle, the detector has a range of $\pm 2\pi$ radians.

Operation of the device may be illustrated by assuming two waveforms, R and V (Figure 1), of the same frequency but differing in phase. If the logic had established by past history that R was leading V, the U output of the detector (pin 4) would produce a positive pulse width equal to the phase difference and the D output (pin 11) would simply remain low.

On the other hand, it is also possible that V was leading R (Figure 1), giving rise to a positive pulse on the D output and a constant low level on the U output pin. Both outputs for the sample condition are valid since the determination of lead or lag is dependent on past edge crossing and initial conditions at start-up. A stable phase-locked loop will result from either condition.

Phase error information is contained in the output duty cycle—that is, the ratio of the output pulse width to total period. By integrating or low-pass filtering the outputs of the detector and shifting the level to accommodate ECL swings, usable analog information for the voltage-controlled oscillator can be developed. A circuit useful for this function is shown in Figure 2.

Proper level shifting is accomplished by differentially

driving the operational amplifier from the normally high outputs of the phase detector ($\bar{U}$ and $\bar{D}$). Using this technique the quiescent differential voltage to the operational amplifier is zero (assuming matched "1" levels from the phase detector). The $\bar{U}$ and $\bar{D}$ outputs are then used to pass along phase information to the operational amplifier. Phase error summing is accomplished through resistors R1 connected to the inputs of the operational amplifier. Some R-C filtering imbedded within the input network (Figure 2) may be very beneficial since the very narrow correctional pulses of the MC12040 would not normally be integrated by the amplifier. General design guides for calculating R1, R2, and C are included in the MC4044 data sheet. Phase detector gain for this configuration is approximately 0.16 volts/radian.

System phase error stems from input offset voltage in the operational amplifier, mismatching of nominally equal resistors, and mismatching of phase detector "high" states between the outputs used for threshold setting and phase measuring. All these effects are reflected in the gain constant. For example, a 16 mV offset voltage in the amplifier would cause an error of $0.016/0.16 = 0.1$ radian or 5.7 degrees of error. Phase error can be trimmed to zero initially by trimming either input offset or one of the threshold resistors (R1 in Figure 2). Phase error over temperature depends on how much the offsetting parameters drift.

FIGURE 1 — TIMING DIAGRAM

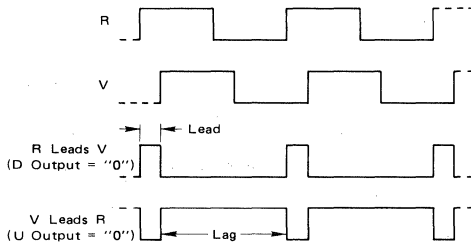


FIGURE 2 — TYPICAL FILTER AND SUMMING NETWORK

